

ACADEMIC DETAILS				
Degree	Specialization	Institute	Year	CPI/%
B.Tech.	Computer Science and Engineering	IIT Gandhinagar	2023-Present	9.25/10
Class XII	Physics, Chemistry, Maths	B. K. D. Aldrich Public School	2021-2022	98.5%
Class X		B. K. D. Aldrich Public School	2019-2020	98%

PUBLICATIONS

• Why Multi-Source Audio Fails to Compose: A Geometric Diagnosis and Its Remedy

[Accepted, BMVC 2026]

(Anurag Singh, Prajwal Singh, Prof. Shanmuganathan Raman)

- Diagnosed compositional failure in multi-source A2I as **geometric displacement** of audio tokens in **CLIP space**, formalized through clustering analysis over intra-class compactness, angular alignment, and inter-token separation.
- Introduced a **mixed inference-time objective** combining **Attend-and-Excite** with an **A* spatial loss**, jointly enforcing token activation and spatial separation of cross-attention maps, resolving neglect and object fusion.
- Redesigned the embedder around a **CLAP** audio encoder with a **multi-token extension (MTE)** per audio source.
- Constructed benchmarks (**VGG2, VGG3, VGGn**) from VGG-Sound and extended **six** A2I metrics to multi-source.
- Achieved upto **2.25×** higher **AIC** over **MACS** and **1.7×** lower **FID** over **AudioToken** with consistent **IIS** gains, ranking best on four of six metrics across **five** benchmarks spanning one to five audio sources.

INTERNSHIPS

• Data Science Intern – Process Intelligence

[May '26 – July '26]

(Skan AI)

- Built a **rule based framework** to detect and filter **noisy events** from raw interaction logs using field and feature level heuristics, tagging each flagged event with an interpretable reason for its classification as noise.
- Designed an **event reordering** module grouping fragmented and interleaved traces into complete **instances** using event fields and noise flags, and explored **VLM based** key value extraction with prompt normalisation.
- Improved the **signal to noise ratio** of captured event logs by filtering diagnosed noise, consistently reducing **process variants** and simplifying the resulting process graphs for downstream process intelligence.

• Research Intern – Compositional Audio-to-Image Generation

[May '25 – July '25]

(Prof. Shanmuganathan Raman, SRIP, IIT Gandhinagar) | [Reports](#)

- Analyzed and benchmarked **20+** models including **AudioToken**, SonicDiffusion, Sound2Scene, and **MACS** on VGG-Sound, **IS3**, and curated datasets for audio-to-image (A2I) generation across six standard evaluation metrics.
- Curated and preprocessed **multi-audio** evaluation sets of **2-5** audio-image pairs from a 50-class VGG-Sound subset.
- Adapted a pretrained AudioToken model to support **multiple audio cues** at inference without additional training.

PROJECTS

• Olympia – Sports Management System

[Jan. '26 – Mar. '26]

(Prof. Yogesh Meena, IIT Gandhinagar) | [Github](#)

- Created a web application using **FastAPI** and **MySQL** with **JWT** authentication, **RBAC**, and session management.
- Performed API endpoint benchmarking with **EXPLAIN analysis** and optimized query performance using **indexing**.
- Implemented **hash based** database **sharding** across multiple tables for horizontal scalability and load distribution.

• Psylang – Custom Programming Language

[Jan. '26 – Present]

(Prof. Shouvik Mondal, IIT Gandhinagar) | [Github](#)

- Designed a language with **Flex** lexer and **LALR(1)** parser supporting closures, lambdas, and user-defined types.
- Built AST-based semantic analyzer enforcing precise scope, type checks, return reachability, diagnostics.
- Implemented TAC IR generator producing quadruples for arithmetic, control flow, functions, short-circuit.

• Ultra-Fast Packet Crafter

[Aug. '25 – Nov. '25]

(Prof. Sameer Kulkarni, IIT Gandhinagar) | [Github](#)

- Modified **Blitzping** to construct and transmit custom **UDP** packets in memory instead of TCP.
- Created a **TTL-based traceroute** using UDP probes and **ICMP** feedback for hop-by-hop path discovery.
- Extended the tool with a **DPDK-based kernel-bypass** for **zero-copy NIC access**, achieving high throughput.
- Outperformed standard Blitzping with **4.5×** **higher packet throughput** and **6.7×** **higher bit-rate transfer**.

- **Custom B+ Tree Database Engine** [Jan. '26 – Mar. '26]
(Prof. Yogesh Meena, IIT Gandhinagar) | [Github](#)
 - Designed and implemented a B+ tree database engine supporting full **CRUD** operations with efficient disk storage.
 - Built a **transaction manager** enforcing **ACID** properties along with **write-ahead logging** system for crash recovery.
 - Implemented schema validation and constraints like **NOT NULL**, **UNIQUE**, **CHECK**, and **foreign key integrity**.
- **Pacman DeepRL Agents** [Aug. '25 – Nov. '25]
(Prof. Manisha Padala, IIT Gandhinagar) | [Github](#)
 - Implemented and compared **5 DRL algorithms** including DQN, DDQN, Deep SARSA, and A2C variants.
 - Trained agents over **400 epochs** using experience replay, target networks, and ϵ -greedy exploration strategies.
 - Improved DDQN with **frame stacking**, enabling temporal awareness and **53% higher peak score**.
 - Implemented A2C with **GAE** and **temperature annealing**, improving stability and convergence performance.
- **Adversarial Robustness of Vision Transformers** [Jan. '25 – Apr. '25]
(Prof. Anirban Dasgupta, IIT Gandhinagar) | [Github](#) | [PyTorch](#), [ViT](#)
 - Evaluated adversarial robustness of **ViT** with sinusoidal, learnable, and RoPE positional embeddings.
 - Proposed a **novel positional encoding** by concatenating positional and patch embeddings in a fixed ratio.
 - Benchmarked over **400 ViT** architectures on CIFAR-10 under black-box attack settings.
 - Achieved **2% accuracy improvement** and outperformed standard positional encodings across attacks.
- **Cycle Sound** [Jan. '25 – Apr. '25]
(Prof. Shanmuganathan Raman, IIT Gandhinagar) | [Github](#) | [ViT](#), [CLIP](#), [VQ-VAE](#), [Transformers](#)
 - Implemented key vision and text-to-vision models including **ViT**, **CLIP**, **VAE**, and **DALL-E v1**.
 - Developed **VQ-VAE** and transformer-based pipelines for audio-to-image and image-to-audio generation.
 - Used a dataset of **24,000+** image-audio pairs from VGG-Sound and performed two-stage training.
- **DAGMM for Anomaly Detection** [Feb. '25 – Apr. '25]
(Prof. Anirban Dasgupta, IIT Gandhinagar) | [Github](#) | [VAE](#), [VQ-VAE](#)
 - Evaluated DAGMM under architectural, loss-function, inference-criterion, and mixture-model modifications.
 - Achieved **1.83%** and **2.24% F1-score** gains by replacing autoencoder with VAE and VQ-VAE in DAGMM.
 - Improved **recall** by **0.19%** while maintaining comparable F1-score through a refined inference criterion.
- **ML Projects** [Aug. '24 – Nov. '24]
[Github-NWP](#) | [Github-HAR](#) | [Decision Trees](#), [TSFEL](#), [LLMs](#)
 - **Next Word Predictor:** Developed a self-supervised text generation model, trained on “War and Peace”, learning word embeddings, and building a custom neural network with adjustable hyperparameters.
 - **Human Activity Recognition:** Built a Human Activity Classifier on the UCI-HAR dataset using decision trees, performed feature extraction with **TSFEL** and employed PCA for result visualization. Also, developed a decision tree from scratch and compared its performance with scikit-learn’s decision tree implementation.

TECHNICAL SKILLS

- **Languages & Libraries:** C, C++, C#, Python, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch.
- **Software & Tools:** Git, GitHub, Xilinx Vivado, Autodesk Inventor, Arduino IDE, $\LaTeX$.

RELEVANT COURSES

- Machine Learning, Deep Learning, Data Science, Data Structures & Algorithms I & II, Foundations of Multi-agent AI, Operating Systems, Computer Networks, Digital Systems, Databases, Compilers, Computer Architecture.

ACHIEVEMENTS

- Earned **Dean’s List** honors in 1st and 2nd semesters (top 5%) and received an **Academic Citation** in the 3rd semester.
- Secured All-India JEE Advanced ranks of **1,544** in 2023 and **2,910** in 2022, placing within the top 2% nationwide.

POSITIONS OF RESPONSIBILITY

- **Executive, Curation Department, TEDxIITGandhinagar** [Aug '24 – May '25]
 - Built a database of **500+** potential speakers, drafted personalized outreach emails, and hosted faculty podcasts.
 - Coordinated speaker logistics including travel, accommodation, and script development.
- **Junior Sponsorship Executive, Amalthea, IIT Gandhinagar** [Aug '23 – Nov '23]
 - Secured sponsorships through proposals, cold outreach, sponsor interaction, and post-event reports.